ORF307: Optimization

Precept: #1

Agenda

- Setting up your Python environment (Codespaces and Docker)
- Linear systems of equations
- Inner and Outer products of vectors
- Matrix multiplication
- Trace of a matrix
- Norms
- Cauchy-Schwartz inequality

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW: ....
- Other announcements on blackboard.

Setting up your Python Environment and Submitting Homeworks

You are free to set up your own Python environment for solving the homework assignments. Once you have completed your work, you must convert it to a **single PDF file** for submission to Gradescope. Please plan your work to submit it ahead of the deadline as extensions will not be given[1].

We suggest two primary methods for setting up your environment:

[1] Given the size of this class, it is not possible to grant extensions (except in rare and extraordinary circumstances). Doing so would significantly delay the release of homework solutions, and impact everyone's ability to prepare for future assignments.

- **GitHub Codespaces**: Detailed instructions are available on the front page of the course companion repository: ORF307/companion.
  A Note on Usage: Through the GitHub Developer Pack, all students receive a free GitHub Pro subscription, which includes 180 core hours and 20 GB of storage per month. Please be mindful of your usage, as continuously running a codespace, even when idle, can consume these hours quickly. To save hours, we strongly recommend that you stop your codespace when you are not actively working.

- **Docker**: Alternatively, you can use the pre-configured environment we have created. Instructions for building the Docker image are available in our dedicated repository: optimization-docker. Additional instructions on docker setup can also be found on Canvas/Modules .
  A Note on Usage: Unlike Codespaces, using the Docker environment on your local machine means you are not restricted by core hours or storage limits. However, the initial setup process may be more complex and time-consuming.

If you have any technical difficulties while setting up your environment, you can always ask the AIs during office hours, or post a question on EdDiscussion.

Linear systems

Linear algebra provides a way to compactly represent linear system

Inner products

Outer products

Matrix multiplication

$A \in \mathbf{R}^{m\times n}$ and $B \in \mathbf{R}^{n\times p}$ then their product $C = AB$ is an $m \times p$ matrix ($C \in \mathbb{R}^{m\times p}$) ...

There are at least 3 different ways to think about matrix multiplication:

- in terms of inner products ...
- in terms of outer products ...
- in terms of matrix-vector products ...

Trace of a matrix

Definition:

Exercise: Show that $Tr(AB) = Tr(BA)$ for all matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$.

Norms

Informally, norms measure the length of a vector $x \in \mathbb{R}^n$. Formally, a norm of a vector $x \in \mathbb{R}^n$ is any function $f : \mathbb{R}^n \to \mathbb{R}$ that satisfies the following **four** properties

1. **Non-negativity**: $f(x) \geq 0, \forall x \in \mathbb{R}^n$
2. **Definitelness**: $f(x) = 0 \iff x = 0$
3. **Homogeneity**: $f(tx) = |t| f(x), \;\; \forall t \in \mathbb{R}, x \in \mathbb{R}^n$
4. **Triangle inequality**: $f(x + y) \leq f(x) + f(y), \;\; \forall x, y \in \mathbb{R}^n$

Examples of norms: The following are the most commonly used norms

- $L2$ or Euclidean norm: ...
- $L1$ norm: ...
- L_∞ norm: ...

Cauchy-Schwartz inequality

It connects the inner product of two vectors with their Euclidean norms

$$x^T y \leq \|x\|_2 \|y\|_2$$